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Atlas of science : visualizing what we know

Börner K., The MIT Press, Cambridge, MA, 2010. 265 pp. Type:
Book (978-0-262014-45-8)

Date Reviewed: Oct 13 2011

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Since 2005, the "Places & Spaces: Mapping Science" exhibit, organized by Indiana University's School of Library and Information Science with support from NSF and other sponsors, has been stimulating a "cross-disciplinary discussion on how to track and communicate human activity and scientific progress on a global scale" (from <http://scimaps.org>). The fundamental idea of "Places and Spaces" is that science, like almost any area of human endeavor, can be represented graphically. Humans are intrinsically spatial beings, and presenting information spatially (a "map") is a natural way to communicate, understand, and manipulate it. The exhibit, which is intended to continue in a variety of venues through 2014, is continually growing. This sumptuous volume is a snapshot of the ideas behind the exhibit as of the 2007 edition.

The book is organized into five parts. Part 1 discusses the exponential rise of science and technology and how graphical representations can help satisfy the needs of people for knowledge. Part 2 traces a history of the graphical display of science. Part 3 outlines a science of science, which can guide this mission of graphical display. Part 4, which is as long as the rest of the book put together, summarizes the first three iterations of the exhibit: the power of maps (the focus of the 2005 edition), the importance of reference systems (the focus of 2006), and the role of forecasts (the focus in 2007). Part 5 anticipates the future of science maps.

With many colorful and intricate graphics and detailed explanatory text, this book will be attractive to a wide range of readers in many different disciplines. The information technology (IT) community should pay attention to it for at least three reasons: it powerfully demonstrates the impact of IT on human knowledge; it is itself a product of IT; and it presents a promising metaphor for future research in IT.

Historically, IT has made possible the exponential growth in knowledge that in turn demands compact representations. Part 1 documents knowledge growth by several metrics, including books, papers, patents, and R&D expenditures, but also human population, which began to explode at the start of the industrial revolution. While the modern digital computer began to be recognizable in the mid-20th century, the industrial revolution had already demonstrated the power of capturing human expertise in machinery, and it is not unreasonable to see common information-theoretic drivers behind the growth of both population and knowledge.

IT not only drives the growth of knowledge and thus the need for knowledge maps, but it also makes possible the collation and detailed visual presentation of large amounts of information. A few of the graphics in the exhibit (including examples from the 15th to the 19th century) were produced before computers, but the vast majority of these displays could never have been produced before the digital computer. The rich variety of visualizations in Parts 2 and 4 will be an ongoing source of inspiration for data analysts in many disciplines.

Finally, the book outlines a way of looking at science (Part 3) that leads to a programmatic outline of how science mapping may develop (Part 5). This provocative discussion deserves careful attention not only by researchers in computer graphics and visualization, but by other computer scientists as well. For example, many algorithmic advances are driven by a topological representation of the domain. This book can be viewed as a guide on how to represent a wide array of domains topologically, opening the door to new advances in manipulating those domains algorithmically.

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